

Ministry Analytics, QA & Live Simulation Engineer

Focus: Ministry Admin Analytics, Chart.js Visualizations, Demo Bot & Test Suite

Executive Summary & Core Responsibility:

Built District & Ministry analytics dashboards, Chart.js throughput trends, live hackathon demo bots, and automated test suites.

1. Technology Stack & Architectural Components

Domain / Layer	Specific Technologies, Libraries & Frameworks Used
Data Visualization	Chart.js 4.x (Dynamic Line Charts, Doughnut Distribution, Bar Load Graphs)
Analytics Aggregation	SQL Analytic Queries (SUM, AVG, Throughput per Hour, Metric Tons, Crores Disbursed)
Bottleneck Diagnostics	SLA Threshold Monitoring (Optimal, Normal, Needs Attention warning badges)
Automated Demo Bot	Auto Demo Script (auto_demo_bot.py) for continuous live presentation playback
QA & Integration Tests	End-to-End Automated Test Suites (test_pipeline.py, test_system.py)

2. Key Modules Developed & Production Deliverables

- **Ministry Admin Command Center (admin_dashboard.html & admin.js):** Built executive dashboard with live KPIs: Total Procured (Metric Tons), Total Disbursed (Crores), Active Vehicles in Queue.
- **Interactive Visualizations:** Developed real-time charts for Hourly Farmer Arrivals vs. Processed Output, Crop Distribution breakdowns, and Mandi Wait Time benchmarks.
- **Bottleneck Detection Table:** Created automated SLA tracking table diagnosing stage delays (e.g. Quality Lab inspection delays) with color-coded alerts.
- **Live Demo Simulation Engine (admin.py & auto_demo_bot.py):** Built 1-click simulation trigger for hackathon presentations to showcase live queue progression automatically.
- **Comprehensive Test Suites:** Developed automated end-to-end regression tests verifying all 5 stage transitions, weight deductions, and API responses.

3. Direct Codebase Ownership & Modified Files

Source Files:

frontend/templates/admin_dashboard.html • frontend/static/js/admin.js • backend/app/routers/admin.py • auto_demo_bot.py • test_pipeline.py • test_system.py

4. SIH Jury Defense & Live Evaluation Points (What to Speak Tomorrow)

- **Presentation Pitch Point:** Demonstrate the Ministry Dashboard showing district-level procurement throughput in real time.
- **Presentation Pitch Point:** Show how the bottleneck detector alerts administrators when weighbridge wait times spike.
- **Presentation Pitch Point:** Trigger the live demo simulation to show judges real-time vehicle flow across all screens.